COUNTING CLICKS, SQUARE-ROOT PRECISION AND RADIOCARBON

How many clicks make a date?

Decay supplies the clock. Logarithms read it. Counting statistics set its precision.

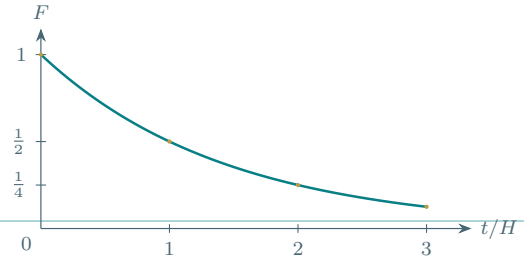
1 Decay and half-life

Activity is $A = \lambda n$, where n is the number of undecayed nuclei. The mean decay law gives

$$n(t) = n_0 e^{-\lambda t}, \qquad A(t) = A_0 e^{-\lambda t}.$$

Writing the half-life as H,

$$H = \frac{\ln 2}{\lambda}, \qquad F = \frac{A}{A_0} = 2^{-t/H}.$$

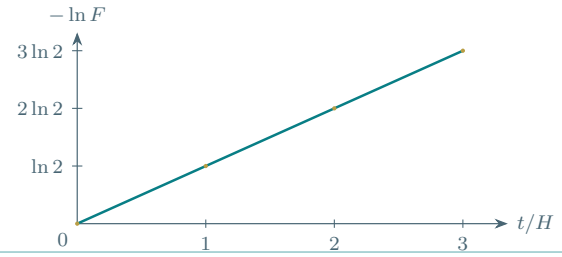


2 Logarithms read the clock

t: elapsed age. T: laboratory counting time.

$$\ln F = -\lambda t, \qquad \boxed{t = -\frac{\ln F}{\lambda}}.$$

For radiocarbon, compare activities for equal amounts of carbon, assuming a closed sample and a known initial activity.



Each half-life adds $\ln 2$ to $-\ln F$.

3 The dice rule returns as counting statistics

As with dice, variances of independent trials add. For M Bernoulli trials, $\mathrm{Var}(N) = Mp(1-p)$. Rare, independent detections give the Poisson limit:

$$N \sim \mathrm{Poisson}(\mu), \qquad \mu = \eta A T, \qquad \sigma_N = \sqrt{\mu} \approx \sqrt{N}.$$

Assume negligible background, known detection efficiency η, and steady activity during $T \ll H$. Using matching time units,

$$\widehat{A} = \frac{N}{\eta T}, \qquad \sigma_A \approx \frac{\sqrt{N}}{\eta T}, \qquad \boxed{\frac{\sigma_A}{\widehat{A}} \approx \frac{1}{\sqrt{N}}}.$$

Absolute count uncertainty grows as $\sqrt{N}$; relative activity uncertainty falls as $1/\sqrt{N}$.

4 Turn an activity error into an age error

$$\sigma_t \approx \left|\frac{\mathrm{d}t}{\mathrm{d}A}\right| \sigma_A = \frac{1}{\lambda}\frac{\sigma_A}{A} \approx \boxed{\frac{1}{\lambda\sqrt{N}}} \propto T^{-1/2}.$$

Here A_0 and λ are treated as known, and enough counts are collected for this local uncertainty approximation.

If an independent reference also has counting uncertainty, with N_0 clicks, then $\sigma_t \approx \frac{1}{\lambda}\sqrt{\frac{1}{N} + \frac{1}{N_0}}$.

5 An idealised radiocarbon age

Take $H = 5730$ years and $F = 1/4$ for a closed sample, with known A_0 and $N = 10\,000$ detected decays.

$$t = \frac{H}{\ln 2} \ln 4 = 11\,460 \text{ years}.$$

$$\sigma_t \approx \frac{5730}{\ln 2\sqrt{10\,000}} \approx 83 \text{ years}.$$

$$\boxed{t \approx 11\,460 \pm 83 \text{ years}}$$

1σ, counting uncertainty only; ideal decay-model age.

Same sample, longer counting

time	clicks	$\sigma_A/\widehat{A}$	σ_t
T_0	10 000	1%	83 yr
$4T_0$	40 000	0.5%	41 yr
$16T_0$	160 000	0.25%	21 yr

Typical counts at the same steady activity. The estimated central age stays the same.

Half the counting error needs four times as long.

6 What laboratory dates also contain

The model above describes decay counting. Modern accelerator mass spectrometry (AMS) measures rare ^{14}C atoms directly. Background, standards and sample preparation also contribute uncertainty.

Conventional reporting uses a corrected fraction modern F and $t_{RC} = -8033 \ln F$ years BP (before AD 1950), based on the 5568-year Libby half-life. Calibration converts this to calendar-age ranges, which can be asymmetric or split.

The $T^{-1/2}$ law describes counting noise; longer counting cannot remove contamination/uncertainty in sample's history.

Counting: NPL (2014), *Practical Radiation Monitoring*, §6.1.2. Radiocarbon: Oxford ORAU, dating and calibration; Historic England, radiocarbon methods.